

$$T.H \quad H_0: z \sim p_0(x) = 1 \{ (0;1) \}$$

$$H_1: z \sim p_1(x) = \frac{e}{e-1} \cdot e^{-x} \{ (0;1) \}$$

a) $n=1$ е. ур. зн. α

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq e$$

$$G: x \leq A$$

$$P(x \leq A | H_0) = \alpha$$

$$\int_0^A 1 dx = A = \alpha$$

$$\alpha_1 = \alpha$$

$$W = P(x \leq A | H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx =$$

$$= \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)} \Big|_0^A = \frac{e}{e-1} (1 - e^{-A})$$

$$\alpha_2 = 1 - W \quad \text{т.к. гипотеза обратная}$$

б) $n=2$ е. ур. зн. α

$$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq e$$

$$G: x_1 + x_2 \leq B$$

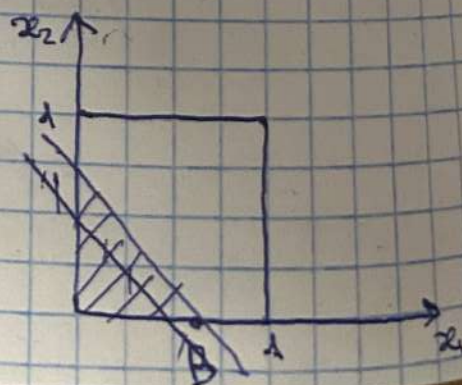
$$P(x_1 + x_2 \leq B | H_0) = \alpha$$

$$\iint 1 \cdot 1 dx_1 dx_2 = \alpha$$

$$x_1 + x_2 \leq B$$

$$0 \leq x_1 \leq 1$$

$$0 \leq x_2 \leq 1$$



$H_0:$

$$B = \frac{1}{2} + \frac{U_2}{\sqrt{12n}}$$

$$G: \bar{x} \leq \frac{1}{2} + \frac{U_2}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B | H_1) \Leftrightarrow$$

$$H_1: M_3 = \int_0^1 x p_1(x) dx = \frac{e-2}{e-1}$$

$$M_3^2 = \int_0^1 x^2 p_1(x) dx = \frac{2e-5}{e-1}$$

$$D_3 = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$\Leftrightarrow P\left(\frac{\bar{x} - M_3}{\sqrt{D_3}} \sqrt{n} \leq \frac{B - M_3}{\sqrt{D_3}} \sqrt{n}\right) = \int_{-\infty}^{\frac{B - M_3}{\sqrt{D_3}} \sqrt{n}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \xrightarrow{n \rightarrow \infty} 1$$

$$f = \frac{\frac{1}{2} + \frac{U_2}{\sqrt{12n}} - 0,418}{\sqrt{D_3}} \sqrt{n} = \frac{U_2}{\sqrt{12}} + 0,082 \sqrt{n} \xrightarrow{n \rightarrow \infty} +\infty$$

Состоятельность критерия:

$$W \xrightarrow{n \rightarrow \infty} 1 \quad \forall H_1 \Rightarrow \text{сост.}$$

$$2) G: x_{\min} \leq C$$

$$P(x_{\min} \leq C | H_0) = \alpha$$

$$H_0: \xi \sim R(0, 1)$$

$$P(x_{\min} \leq C) = 1 - (1 - F(c))^n = \alpha$$

$$C = 1 - \sqrt[n]{1 - \alpha}$$

$$G: x_{\min} \leq 1 - \sqrt[n]{1 - \alpha}$$

$$\alpha = \alpha$$

$$W = P(x_{\min} \leq C | H_1) = 1 - (1 - F(c))^n =$$

12. У игроков
впечатление,
остальные две гра.

$$= \int_0^{\infty} F_1(x) = \int_0^{\infty} \frac{e}{e-1} e^{-x} dx = \frac{e}{e-1} (1 - e^{-1+\sqrt{1-\alpha}}) =$$

$$= 1 - \left(1 - \frac{e}{e-1} (1 - e^{\sqrt{1-\alpha}-1}) \right)^n$$

< 1

$\xrightarrow{n \rightarrow \infty} 1 \Rightarrow$
 $\Rightarrow \text{сост.}$

$\alpha_2 = 1 - W$

$T_{12} \quad \alpha = 0,2$

$H_0: \{ \sim p_0(x) = \frac{1}{4}\{1\} + \frac{1}{4}\{2\} + \frac{1}{6}\{3\} + \frac{1}{3}\{4\}$

$H_1: \{ \sim p_1(x) = \frac{1}{4}\{1\} + \frac{1}{4}\{2\} + \frac{1}{4}\{3\} + \frac{1}{4}\{4\}$

$n = 2$

$$L = \frac{p_1(x_1) \cdot p_1(x_2)}{p_0(x_1) \cdot p_0(x_2)}$$

1 \ 2	1	2	3	4
1	1	1	$\frac{3}{2}$	$\frac{3}{4}$
2	1	1	$\frac{3}{2}$	$\frac{3}{4}$
3	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{9}{4}$	$\frac{9}{8}$
4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{9}{8}$	$\frac{9}{16}$

H_0

1 \ 2	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

H_1

1 \ 2	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

$$G: l \geq c$$

$$P(l \geq c | H_0) \leq 0,2$$

Начинаем с $\frac{9}{4} (3;3)$, потом возводим
и разд по $\frac{3}{2}$

получаем $0,1944 \leq 0,2$ (Большее не
получится)

$$G = \{(3;3), (1;3), (3;1), (2;3), (3;2)\}$$

$$\alpha_1 = \frac{7}{36} - \text{ошибка первого рода}$$

$$W = \frac{5}{16} \Rightarrow \alpha_2 = \frac{11}{16}$$

↙ все исходы равновероятны